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An Empirical Study of Scaling Law for Scene Text Recognition

Individual Work: Pattern Recognition Innovations

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1 Overview Description of the Selected Solution

Link to Paper: [An Empirical Study of Scaling Law for Scene Text Recognition \(CVPR 2024\)](#)

1.1 Which problem does this solution address?

This solution addresses two critical gaps in the field of Scene Text Recognition (STR):

1. **The lack of empirical investigation into Scaling Laws:** Although relationships between model size, data, and performance are well-established in NLP, they had not been explored in STR. Prior research focused on architectural tweaks with fixed datasets, creating uncertainty about whether simply "scaling up" would work.
2. **The Data Composition Problem:** There was no clear guideline on how to effectively combine real-world and synthetic data. This study resolves this by systematically quantifying the impact of data mixtures, identifying the optimal ratio (specifically 1:3) to maximize recognition accuracy.

By solving these issues, the study provides a principled guideline for designing future high-performance OCR systems.

1.2 Who develops this solution, and who are potential users?

Developers: The solution was developed by researchers from **Huawei Noah's Ark Lab**, a leading AI research institute. The primary authors include Miao Rang, Zhenni Bi, and colleagues.

Potential Users: The main users include researchers in deep learning and computer vision who study large-scale model behavior and performance trends. In addition, developers of commercial OCR systems such as those applied in autonomous driving, street-view analysis, and large-scale document digitization can directly benefit from these findings to improve both training efficiency and recognition accuracy.

1.3 A Brief Development History

This work represents a paradigm shift in STR research from architectural innovation toward scale-driven optimization.

- **Foundation:** The study builds upon the success of Transformer-based models such as *TrOCR* and *PARSeq*, which previously achieved state-of-the-art performance by leveraging pre-trained vision and language models.
- **Innovation (2024):** Published at *CVPR 2024*, the authors moved beyond fixed-scale experiments by constructing a large-scale dataset called *REBU-Syn*, consisting of 6 million real-world samples and 18 million synthetic samples. Models ranging from 22 million to 1 billion parameters were trained to study scaling behavior.
- **Achievement:** The scaling strategy led to the development of *CLIP4STR-L*, which achieved a new state-of-the-art top-1 average accuracy of 97.42% across six standard benchmark datasets.

1.4 Evidence (References)

- **Scientific Publication:** Published in the proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
- The authors have released the source code and the REBU-Syn dataset statistics on their [project homepage](#), allowing the scientific community to reproduce the scaling laws.
- The authors provide reproducible code and detailed statistics of the *REBU-Syn* dataset to support the validity of the proposed solution: [GitHub](#).

2 Key Features and Pattern Recognition Functions

2.1 Smooth Power Law Modeling

The central contribution is the discovery that recognition error follows a smooth power-law relationship with respect to computational resources (model parameters, data size, and compute).

- **Pattern Recognition Function:** The system recognizes that model performance improves in a predictable, linear pattern on a log-log scale. This relationship is governed by a power-law equation (detailed in the Reference), where the error rate decreases as the scale increases.
- **Significance:** This mathematical predictability implies that visual text recognition does not saturate early. It allows researchers to accurately forecast the performance of billion-parameter models by extrapolating the pattern from smaller experimental runs, without needing to train them fully first.

2.2 Sample Efficiency in Pattern Learning

The study shows that large-scale models are significantly more sample-efficient than smaller ones.

- **Function:** When exposed to the exact same number of training images, a larger model (e.g., 1 Billion parameters) recognizes text patterns more accurately than a smaller model.
- **Algorithm:** The solution primarily utilizes the **TrOCR** architecture (Transformer-based Optical Character Recognition) with a Vision Transformer (ViT) encoder. These algorithms use Self-Attention mechanisms to capture global dependencies across the entire image, allowing the model to utilize context from the whole word to predict individual characters accurately.

2.3 Large-Scale Dataset Integration (REBU-Syn)

To enable this pattern recognition capability, the authors constructed *REBU-Syn*, a massive consolidated dataset.

- **Composition:** It combines **6 million real-world images** (from datasets like Uber-Text, COCO-Text) with **18 million synthetic samples**.
- **Significance:** This establishment of a new data standard is crucial for training high-performance pattern recognizers, as the "Scaling Law" dictates that model growth must be accompanied by data growth.

3 Evaluation: Pros, Cons, and Future Potential

3.1 Pros

- **Predictability:** The power-law formulation allows researchers to estimate the computational budget required to reach a specific accuracy target, eliminating trial-and-error.
- **State-of-the-Art Performance:** Achieved an average accuracy of 97.42% on standard benchmarks, outperforming all previous specialized architectures.
- **Optimized Data Strategy:** The study discovered that the optimal training mix is a **1:3 ratio** of real to synthetic data, providing a concrete guideline for future dataset creation

3.2 Cons

- **High Computational Cost:** Validating these laws required training models with up to **1 billion parameters**, which demands expensive GPU clusters and substantial energy consumption.
- **Diminishing Returns:** The study notes that as accuracy approaches 100%, the amount of data/compute required to gain an additional 0.1% improvement grows exponentially.

3.3 Future Potential

This research lays the groundwork for **OCR Foundation Models** — universal models capable of reading any text in any language without retraining. Future applications may extend these scaling laws to handwriting recognition and historical manuscript analysis.

4 Similar Solutions

4.1 Comparison with Predecessor: PARSeq_A[#]

- **Quantitative Comparison:** According to the results (Figure 4 below), PARSeq_A[#] achieves an average accuracy of **95.57%**. In contrast, the scaled model CLIP4STR-L achieves **97.42%**.
- **Analysis:** The substantial improvement of **+1.85%** confirms that scaling up the data and model size (the "Scaling Law" approach) is more effective than relying solely on complex architectural modifications like the permuted autoregressive mechanism used in PARSeq.

4.2 Comparison with Competitor: ABINet

- **Quantitative Comparison:** ABINet achieves an accuracy of approximately **96.01%**. In contrast, the scaled model CLIP4STR-L outperforms it with **97.42%** accuracy.
- **Analysis:** This result is significant because ABINet relies on complex, heavy linguistic reasoning to fix errors. The fact that CLIP4STR-L wins purely through **visual scaling** (increasing model size and data) indicates that a sufficiently large visual model can learn language properties implicitly without needing a separate, explicit language module.

Reference

A. Formula Insertion (For Section 2.1 - Smooth Power Law Modeling)

The formula is expressed as:

$$E = aC^b$$

Where the constants a and b are derived empirically by fitting the training results of smaller models (S, B, L) to predict the behavior of larger ones.

Note: Model performance E is primarily influenced by three variables: the number of model parameters N , the volume of the training data D , and the computation of the model C .

The power law of model when data is fixed:

- Scaling TrOCR Models:

$$E(N) = \left(\frac{1.97 \times 10^4}{N} \right)^{0.223}$$

The power-law of data when model size is fixed:

- Scaling Data Volume on TrOCR:

$$E(D) = \left(\frac{1.84 \times 10^5}{D} \right)^{-0.327}$$

The power law of computation:

$$E(C) = \left(\frac{4.45 \times 10^4}{C} \right)^{-0.3271}$$

B. Reference 1: The Scaling Law Graph

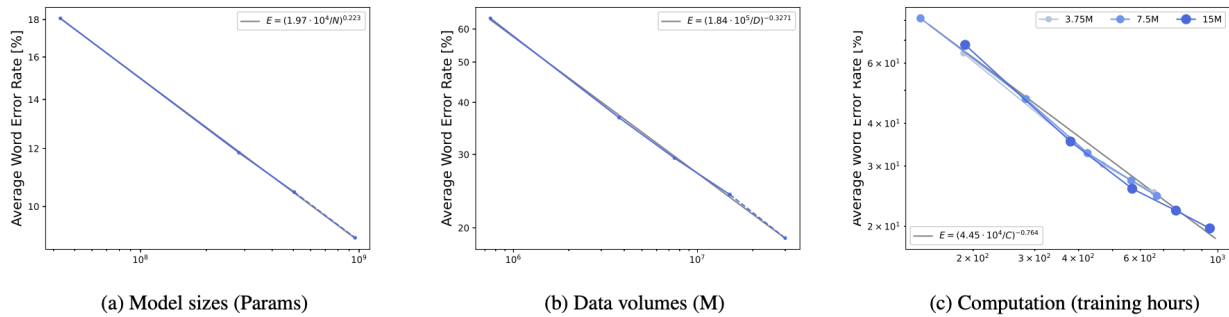


Figure 1: **Empirical Scaling Laws in STR.** The experimental results demonstrate a consistent power-law relationship where the recognition error rate decreases linearly on a log-log scale when scaling up three key factors: (a) Model Size (Parameters), (b) Data Volume, and (c) Computational Budget (Training Hours).

C. Reference 2: Proportion of Training Data

Real DataSet	Syn DataSet	Data Ratio	Word Acc
R+E+B+U	Syn	1:0.5	96.19
R+E+B+U	MJ+ST	1:2.5	96.24
R+E+B+U	MJ+ST+Syn	1:3	96.85

Figure 2: PARSeq-S average accuracy of integrating diverse synthetic and real data types.

Data Ratio	Word Acc
Real:Syn=1:0.5	96.32
Real:Syn=1:1	96.50
Real:Syn=1:2	96.59
Real:Syn=1:3	96.85
Real:Syn=1:4	96.76
Real:Syn=1:5	95.70

Figure 3: PARSeq-S average accuracy on 6 common test benchmarks with varying ratios of synthetic and real data.

D. Reference 3: Performance Comparison Table

Method	Training data	Regular Text			Irregular Text			Avg
		IC13 857	IIT5k 3,000	SVT 647	CUTE80 288	IC15 1,811	SVTP 645	
ViTSTR-B [1]	MJ+ST	92.40	88.40	87.70	81.30	72.60	81.80	85.46
SE-ASTER [44]	MJ+ST	92.80	93.80	89.60	83.60	80.0	81.40	88.35
TRBA [3]	MJ+ST	93.4	92.10	88.90	89.20	77.4	89.30	90.07
SRN [69]	MJ+ST	95.5	94.80	91.50	87.80	82.70	85.10	90.42
TextScanner [61]	MJ+ST	94.90	95.70	92.70	91.60	83.50	84.80	91.15
VisionLAN [65]	MJ+ST	95.7	95.80	91.70	88.50	83.7	86.00	91.23
PETR [66]	MJ+ST	97.00	95.80	92.40	89.90	83.30	86.20	91.42
ABINet [16]	MJ+ST	97.4	96.20	93.50	89.20	86.00	89.30	92.66
MaskOCR(ViT-B) [37]	MJ+ST	98.1	95.8	94.7	89.2	87.3	89.9	93.1
PARSeq _A [6]	MJ+ST	96.20	97.00	93.60	92.20	82.90	88.90	93.16
MATR _N [39]	MJ+ST	95.80	96.60	95.00	93.50	82.80	90.60	93.20
TrOCR _{Large} [31]	MJ+ST+B	97.30	94.10	96.10	95.10	84.10	93.00	93.23
DiG-ViT-B [68]	MJ+ST	96.90	96.70	94.60	91.30	87.10	91.00	93.41
MaskOCR(ViT-L) [37]	MJ+ST	98.2	98.0	96.9	95.8	90.1	94.6	93.78
ViTSTR-S [1]	Real	97.80	97.90	96.00	96.20	89.00	91.50	94.85
DiG-ViT-B [68]	Real	97.60	97.60	96.50	96.50	88.90	92.90	94.86
PARSeq _A [#] [6]	Real	97.32	97.87	97.37	97.22	90.34	94.73	95.57
ABINet [16]	Real	98.00	98.60	98.20	97.20	90.50	94.10	96.01
MAERec-B [23]	Union14M-L	98.10	98.50	97.80	98.60	90.7	94.40	96.15
CLIP4STR-B [73]	Real	98.36	98.73	97.68	98.96	91.39	96.74	96.89
CLIP4STR-L [73]	Real	98.48	99.43	98.15	98.96	91.66	97.36	97.06
CLIP4STR-B*	REBU-Syn	99.29	98.96	98.76	99.65	92.27	97.83	97.25
CLIP4STR-L*	REBU-Syn	99.42	99.13	98.61	99.65	92.6	98.13	97.42

Figure 4: Comparison with state-of-the-art methods. The scaled model (CLIP4STR-L) outperforms previous competitors like ABINet and PARSeq across all six benchmarks.